Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech (IT) Examination

April 2018

IT212 Computer Organization & Microprocessor

Date: 30.04.2018, Monday **Time: 10.00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 **Answer the following questions.**

a. How negative numbers are represented in computer? What are the advantages of the representation? **[03]**

b. Explain the various special purpose register available in MIPS processor. **[04]**

Q – 2.a Explain the conditional control transfer instructions. **[04]**

Q – 2.b What is an overflow? Explain the overflow for the addition using suitable example. **[04]**

Q – 2.c Explain the various addressing modes followed in MIPS. **[06]**

OR

Q – 2.a What is sign extension? Explain why it is essential in arithmetic operation. **[04]**

Q – 2.b Explain OR, AND, NOT and EXOR operation using suitable example. **[04]**

Q – 2.c Explain each field of R-Type instruction format. **[06]**

Q – 3.a What is JAL instruction? List the various things, which need to be pushed during the method call? Which general-purpose register is used to handle method call? **[06]**

Q – 3.b **Convert following code to equivalent MIPS code: (Any two)** **[08]**

• While (i>3) { • a = a + 12345678 • b = b ^ 2; • i++; • }	• for(i=10; i > j; --i) { • k = k * 2 ; • i = i / 2 ; • }	• If (a = = b) • b = c and 2 • else if (c = = d) • d = NOT d

SECTION – II

Q - 4 **Answer the following questions.** **[07]**

a. Convert -5.789 into binary.

Q – 5.a Explain the working of booth multiplication using suitable example. **[07]**

OR

Q – 5.a	List the steps to perform division operation over the binary data.	**[07]**
Q – 5.b	Explain set-associative cache using suitable example.	**[07]**

OR

Q – 5.b	Explain fully associative cache using suitable example.	**[07]**
Q – 6.a	Write the difference between RISC and CISC.	**[04]**
Q – 6.b	How the performance of computer can be measured?	**[05]**
Q – 6.c	Draw the Single Cycle data path for "SUB" instruction.	**[05]**

OR

Q – 6.a	Write the MIPS code to add 10 elements of the array.	**[04]**
Q – 6.b	Explain the pipeline using suitable example.	**[05]**
Q – 6.c	What is virtual memory? What are the advantage of having virtual memory in the processor?	**[05]**
